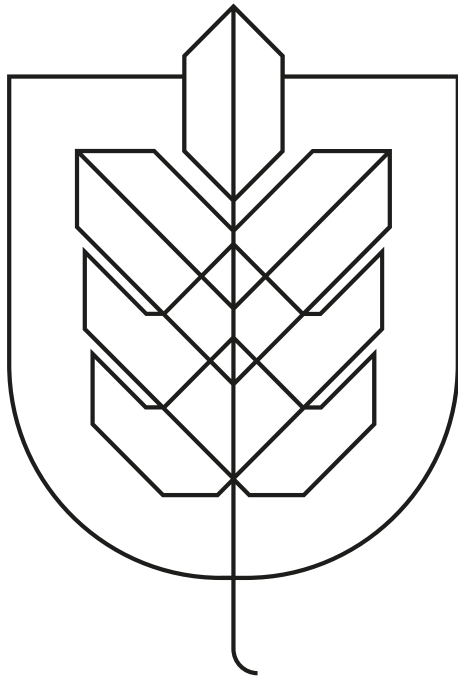




HULT
INTERNATIONAL
BUSINESS SCHOOL

A3: Individual Assignment

Pneumonia Classification Using CNN, VGG16 & DenseNet121

Introduction to Machine Learning

Sunday, 20th of April 2025

By: Habib Mohamed Sultan Saleem

MSc Business Analytics (MBAN)

Table of Contents

A3: DESIGN & IMPLEMENTATION OF E2E PIPELINE USING SNOWFLAKE AND TABLEAU	1
INTRODUCTION AND PROBLEM DEFINITION.....	3
DATASET CONTEXT AND DESCRIPTION.....	3
ALGORITHM SELECTION AND JUSTIFICATION	4
CODE EXPLANATION AND IMPLEMENTATION.....	5
DOCUMENTATION OF ENSEMBLE TECHNIQUES USED	6
PERFORMANCE IMPROVEMENTS IMPLEMENTED	7
AI TOOL USAGE DISCLOSURE	10
CONCLUSION	10
BIBLIOGRAPHY	11

Introduction and Problem Definition

Pneumonia is a serious respiratory infection that can be life-threatening. Pneumonia can be caused by a range of pathogens, such as bacteria, viruses, or fungi, thus requiring timely diagnosis to enable appropriate treatment and reduce mortality. Chest X-rays are a key feature of pneumonia diagnosis, especially in paediatric cases where early identification can prevent additional complications. However, X-ray image interpretation often involves subjectivity, requires a significant amount of time, and relies on skilled radiologists, whose availability can be limited.

Around the world, particularly in developing nations, pneumonia remains the leading cause of death among children under the age of five. This project aims to solve this crucial limitation through creating an ML tool designed to automatically classify chest X-ray images.

Our aim is to classify chest X-ray images into three categories of normal, bacterial and viral pneumonia. Automating classification processes using ML techniques can significantly improve efficiency of diagnostics, reduce dependencies on radiologists, and aid hospitals facing heavy patient loads or limited radiologic resources. Using supervised learning methods to solve this problem, we intend to train models using labelled data, enabling enhanced decision-making processes in real-world clinical settings.

Dataset Context and Description

We created and tested this model using an open-source, real-world dataset from Guangzhou Women and Children's Medical Centre which includes anterior-posterior chest X-ray images of subjects aged between 1 to 5 years. Images were systematically acquired under routine clinical practice, labelled under expert radiologist interpretation.

Data is segregated into three high-level directories i.e., train, validation, and test, each containing subdirectories corresponding to respective classes. Two classes existed in initial stages of the dataset, like NORMAL and PNEUMONIA. To separate the reason behind cause of pneumonia more clearly, we further subdivided class PNEUMONIA into BACTERIAL and VIRAL based on filename metadata and visual inspection, thus introducing the third class, which imparted an added element of granularity along with real-world complexity to the classification process.

Facts about the dataset:

Found 4173 images belonging to 3 classes.
Found 1043 images belonging to 3 classes.
Found 624 images belonging to 3 classes.

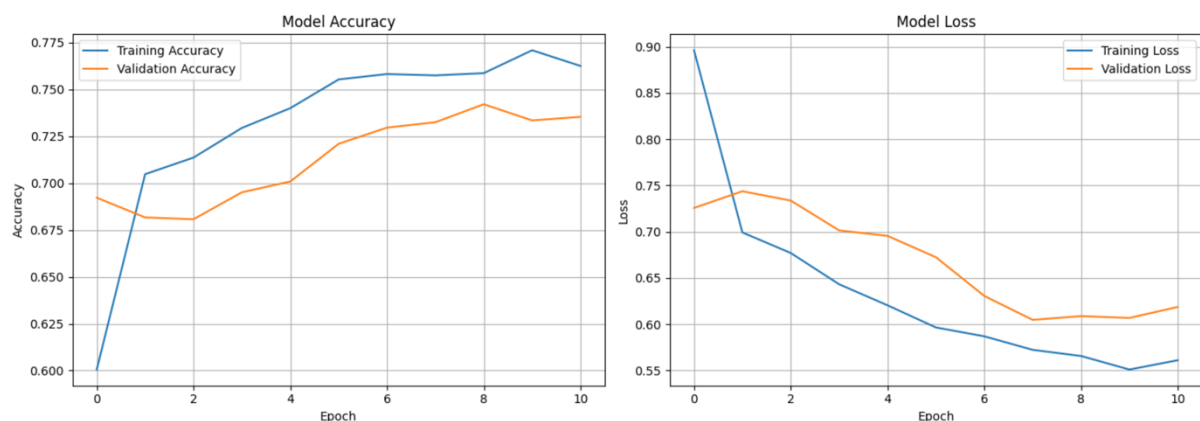
- Size: 5,840 chest X-ray images in JPEG format
- Image type: Anterior-posterior X-rays

- Labels verified by expert radiologists
- Structure: Three folders such as train, test, val each with class-based subfolders

This dataset has several challenges such as class imbalance with fewer viral pneumonia cases and visual similarity between Normal, bacterial and viral pneumonia which makes the classification difficult.

Algorithm Selection and Justification

This project addresses traditional supervised learning classification problem, where the goal is to formulate a model that utilizes labelled data to classify unseen chest X-ray images into one of three categories, namely, Normal, Bacterial Pneumonia, or Viral Pneumonia. Training process is regulated towards reducing categorical cross-entropy loss using the process of backpropagation, which enables the model to learn from mistakes experienced during training time. During this research, we explored three separate deep learning approaches, including a custom-made Convolutional Neural Network (CNN) model built from scratch, as well as two transfer learning models based on pretrained architectures such as VGG16 and DenseNet121.



First model is CNN, a deep learning architecture for image classification. It learns spatial hierarchies of features such as edges and shapes using convolutional layers, interleaved with max pooling to reduce dimensionality. CNN model showed signs of overfitting due to relatively small dataset.

To solve this problem, we utilized transfer learning with VGG16, a model pre-trained using ImageNet. Because of its deep layers and learned weights through training, VGG16 is skilled at feature extraction of images of many different kinds. Freezing convolutional base and only training higher-level layers, we created a classifier specifically designed with pneumonia detection, which provided a lower number of trainable parameters. This strategy improved performance at generalization.

To further explore, we integrated DenseNet121, which is a pretrained architecture that is used exclusively for medical image analysis. Unlike VGG16, DenseNet is densely connected, meaning that each layer receives input from all previous layers, which allows for deeper gradient flow. These features make DenseNet suitable for identifying subtle patterns in complex medical images.

DenseNet has been successfully applied to real-world healthcare environments, specifically to examination of chest radiographs and abnormal finding of pneumonia, tuberculosis, and COVID-19. In our study, while DenseNet was not bested overall by VGG16, it performed well at detecting bacterial cases.

Finally, we formed a soft voting ensemble through computation of mean of the predicted class probabilities from CNN, VGG16, and DenseNet architectures. By doing so, we aimed to reduce prediction variance and counter biases of the individual models.

Code Explanation and Implementation

Codes were made with Python using libraries like TensorFlow and Keras, ran inside Jupyter Notebook application. Data loading and preprocessing composed the first part of the project, using ImageDataGenerator class to handle normalization, along with data augmentation techniques like shearing, horizontal flipping, zooming, and slight rotations. These transformations helped in simulating realistic clinical variations due to small dataset by improving diversity of images in training set. Data was structured systematically into subdirectories corresponding to each class, thus allowing for automatic labelling based on folder names.

```
def build_baseline_cnn(input_shape=(150, 150, 3), num_classes=3):
    model = Sequential()

    # Add input layer
    model.add(Input(shape=input_shape))

    # First convolutional block
    model.add(Conv2D(32, (3, 3), activation='relu'))
    model.add(MaxPooling2D(pool_size=(2, 2)))

    # Second convolutional block
    model.add(Conv2D(64, (3, 3), activation='relu'))
    model.add(MaxPooling2D(pool_size=(2, 2)))

    # Third convolutional block
    model.add(Conv2D(128, (3, 3), activation='relu'))
    model.add(MaxPooling2D(pool_size=(2, 2)))

    # Flatten and Dense layers
    model.add(Flatten())
    model.add(Dense(128, activation='relu'))
    model.add(Dropout(0.5))

    # Output layer with softmax for 3-class classification
    model.add(Dense(num_classes, activation='softmax'))

    # Compile the model
    model.compile(
        optimizer='adam',
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )

    return model
# Part of this code was generated with the assistance of ChatGPT
```

First model used was CNN intended to be baseline to which others would be compared. Network architecture involved three convolutional layers using ReLU activation functions, followed by MaxPooling2D layers that successively decreased spatial dimensionality while preserving important features. Feature mapping was transformed into one-dimensional vectors

flattening operation, which then passed through dense hidden layer, where dropout was added to counteract danger of too much regularization, before reaching final softmax output classifying into three classes classification. Model being trained by the Adam optimizer using categorical cross-entropy loss function, training regime was tracked using the metric of validation accuracy.

```
# Add custom layers on top of VGG16
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(128, activation='relu')(x)
x = Dropout(0.5)(x)
output = Dense(3, activation='softmax')(x)

# Define the final model
vgg_model = Model(inputs=base_model.input, outputs=output)

# Compile the model
vgg_model.compile(
    optimizer=Adam(learning_rate=0.0001),
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
# Part of this code was generated with the assistance of ChatGPT
```

To improve baseline performance, transfer learning was leveraged using VGG16 model, which has previously been pre-trained using ImageNet. A pretrained convolutional base used the `include_top=False` to hide the initial classifier head. A new classification head consisting of `GlobalAveragePooling2D`, one dense layer with 128 units using ReLU activation, dropout, and concluding with the final softmax output layer targeted to classify the three separate classes of pneumonia was then added. The base layers of the model remained frozen, thus limiting the training to the newly added layers only. This approach largely reduced trainable parameters, thus improving the model's computational effectiveness using pretrained characteristics.

Further building a DenseNet121 model as an enhanced transfer learning model. The unique feature of DenseNet is architecture, which supports dense connections between layers, with each receiving inputs from all previous layers, thus aiding efficient feature reuse and gradient propagation. The convolutional base was kept fixed, like VGG16 architecture, with addition of an individualized classification head. DenseNet121 was effective at feature extraction showing potential to distinguish subtle differences between radiologic images based on X-rays however, the model showed indications of overfitting, indicated by enhanced training accuracy set against decreased validation performance after multiple epochs, implying that the model is too sophisticated compared to dataset size.

Ultimately, to enhance robustness of predictions and take full use of the strengths inherent to different models, an ensemble learning strategy was adopted. Predictions made by baseline CNN model, along with VGG16 and DenseNet121 model, were combined using soft voting, which involves calculating the mean of class probabilities predicted by each individual model and then taking the class with the highest mean probability.

Documentation of Ensemble Techniques Used

Ensemble Classification Report:				
	precision	recall	f1-score	support
BACTERIA	0.37	0.58	0.45	242
NORMAL	0.36	0.30	0.33	234
VIRUS	0.30	0.09	0.14	148
accuracy			0.36	624
macro avg	0.34	0.33	0.31	624
weighted avg	0.35	0.36	0.33	624

To enhance classification performance and reduce variability between individual models, an ensemble learning model was utilized. Ensemble learning is a paradigm where several models are combined to produce predictions that are more stable and accurate than those from one individual model independently. This method is specifically beneficial where precision of the decisions is highly crucial, like in medical imaging.

In this study, we used softvoting ensemble approach. This process involves taking the mean of the class probabilities generated by each model and mapping the class with the highest mean probability to final response. Unlike hard voting, which does not take into consideration the confidence scores accompanying each prediction. Softvoting makes it more sensitive to subtle differences between model predictions.

These predictions came from three independent deep learning frameworks:

- Baseline CNN, trained from scratch on pneumonia dataset.
- VGG16 transfer learning model employs an existing convolutional base that is complemented with customized dense layers.
- DenseNet121 transfer learning architecture utilizes dense connections to facilitate feature re-use process, as well as gradient propagation.

Class probabilities for test data were produced by each model via the use of the softmax activation function on the output layers relevant to each target model. These probabilities were then combined across the three models to create a unified set of predictions.

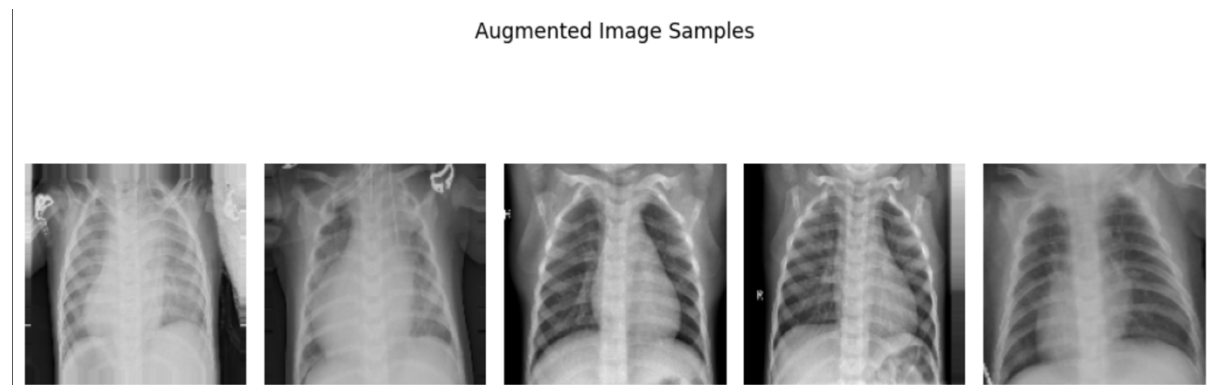
The resulting ensemble model achieved an accuracy rate of 36% and showed improved effectiveness, particularly in Bacterial Pneumonia class. Though improvement in overall accuracy was small, the ensemble showed more stability in predictions made, reflecting an efficient understanding of ML concepts relevant to practical medical diagnostic scenarios.

This combined strategy highlighted potential to use a wide collection of model architectures, ranging from shallow, specialized convolutional neural networks to large pre-trained models, to improve resilience with only modest extra computational cost. It highlights the value of ensemble learning, not only to improve performance, but to improve reliability where it matters, such as the field of medicine.

Performance Improvements Implemented

Several methodologies were utilized to improve performance of model in successive iterations, including measures to address problems like small dataset size, overfitting, class imbalance, and poor generalization performance to test dataset.

Data Augmentation



To counter restrictions caused by the dataset's comparatively small size, and enhance the model's generalisation, real-time data augmentation was carried out using ImageDataGenerator class. The transformations included flipping, zoom, shear, and rotation, mimicking variability present in real-world settings.

Classification Report:				
	precision	recall	f1-score	support
BACTERIA	0.41	0.57	0.48	242
NORMAL	0.43	0.33	0.38	234
VIRUS	0.27	0.20	0.23	148
accuracy			0.39	624
macro avg	0.37	0.37	0.36	624
weighted avg	0.39	0.39	0.38	624

The result of augmentation was noticeable in CNN model despite its simple architecture, the CNN model achieved an accuracy score of 39% and a macro F1-score of 0.36.

Using pre-trained models for Transfer Learning

We employed two transfer learning methods using VGG16 and DenseNet121, which are pre-trained using ImageNet. By keeping the convolutional backbone of the architectures fixed and adding customized classification heads, we kept the strong, feature extraction capabilities intact while targeting training at the higher-level layers that are specifically crafted to classify

pneumonia. This lowered the number of trainable parameters, and the time needed to train, improving the model's generalizability to unseen, unobserved data.

Classification Report:				
	precision	recall	f1-score	support
BACTERIA	0.38	0.50	0.44	242
NORMAL	0.39	0.40	0.39	234
VIRUS	0.29	0.14	0.18	148
accuracy			0.38	624
macro avg	0.36	0.35	0.34	624
weighted avg	0.36	0.38	0.36	624

VGG16 Classification Report

Classification Report:				
	precision	recall	f1-score	support
BACTERIA	0.39	0.48	0.43	242
NORMAL	0.36	0.33	0.34	234
VIRUS	0.27	0.20	0.23	148
accuracy			0.36	624
macro avg	0.34	0.34	0.34	624
weighted avg	0.35	0.36	0.35	624

DenseNet121 Test Evaluation Summary
DenseNet121 Classification Report

VGG16 model, using the feature extraction mode, did not achieve better results than the other CNN model, with an accuracy score of 38% and a macro F1-score of 0.34. VGG16 specifically showed aptitude in classifying BACTERIA samples, with an F1-score of 0.44, and NORMAL samples, with an F1-score of 0.39, but struggled to classify the VIRUS category correctly, with an F1-score of 0.18.

DenseNet121 took a different route toward transfer learning, reaching an equivalent test accuracy of 36% with a macro F1-score of 0.34. Though not better than VGG16 and the CNN, DenseNet proved more stable throughout training process and returned more consistent results per validation fold.

DenseNet121 for Deep Feature Learning

DenseNet121 was created to improve overall performance. Dense blocks in DenseNet allow efficient passing of the gradients and parameter sharing, aiding the model to grasp subtle patterns of chest X-ray images, particularly helpful in distinguishing fine differences between viral and bacterial pneumonia.

Although performance gains were incremental, DenseNet exhibited increased stability during the training iterations and obtained better validation accuracy compared to baseline model. Specifically, DenseNet121 recorded a recall of 0.48 for the BACTERIA class and showed increased resistance to overfitting compared to CNN.

Early Stopping and Model Checkpointing

```
# Stop training if val_loss doesn't improve for 3 epochs
early_stop = EarlyStopping(
    monitor='val_loss',
    patience=3,
    restore_best_weights=True
)

# Save the best model weights during training
checkpoint = ModelCheckpoint(
    'best_cnn_model.h5',
    monitor='val_loss',
    save_best_only=True
)

# Group callbacks
callbacks = [early_stop, checkpoint]
```

To ensure training without overfitting, EarlyStopping and ModelCheckpoint callbacks were used. Early stopping observed the validation loss and stopped training when no improvement was observed over epochs, preventing model from overfitting. ModelCheckpoint saved the best-performing model based on validation accuracy, ensuring final model used was effective.

AI Tool Usage Disclosure

Throughout project, AI tools such as ChatGPT, were utilized in an ethically responsible manner. Support was limited to specific tasks, such as the correction of coding mistakes, explanation of syntax, and provision of conceptual advice. On several occasions, these tools further aided the study of topics not covered under the current curriculum, such as transfer learning via the use of pre-trained models such as VGG16 or DenseNet121, model checkpointing and early stopping. AI tools became an educational tool to supplement the knowledge gap. As emphasized by Dr. Salter a computer science technologist and entrepreneur with a Ph.D. in Robotics the use of AI in coding is encouraged when used responsibly. She advocates for leveraging AI to boost efficiency while ensuring that developers grasp the core logic behind their implementations. This practice transforms AI into a learning partner rather than a shortcut, reinforcing conceptual understanding and professional growth.

I used AI to grasp intrinsic principles behind each approach or operation before independently wrote the code. Each implementation was created from scratch, founded upon careful examination, input-output relationships, and expected performance. The final code was run under Jupyter Notebook, carefully screened for validation, and documented. Whenever artificial intelligence support was utilized, it was analyzed, modified, and optimized fully to align with the project goals and design. This ensures that all work submitted reflects my understanding, thus ensuring academic integrity.

Conclusion

This project successfully implemented a complete ML pipeline for pneumonia detection using chest X-ray images. Starting with a baseline CNN, we progressed to transfer learning with VGG16 and DenseNet121. CNN delivered best individual performance with 39% accuracy even though it still requires more fine-tuning. DenseNet provided an improved validation consistency but showed signs of overfitting. Finally, an ensemble of all three models using soft voting demonstrated an advanced ML application. We applied key concepts such as data augmentation, regularization, and model evaluation, aligning with real-world ML practices in the medical image classification.

Bibliography

Ibm. (2025, April 15). What are convolutional neural networks? *IBM*.

<https://www.ibm.com/think/topics/convolutional-neural-networks>

Great Learning. (2022, January 5). Everything you need to know about VGG16 - Great

Learning - Medium. *Medium*. <https://medium.com/@mygreatlearning/everything-you-need-to-know-about-vgg16-7315defb5918>

Zhou, T., Ye, X., Lu, H., Zheng, X., Qiu, S., & Liu, Y. (2022). Dense convolutional network and its application in medical image analysis. *BioMed Research International*, 2022, 1–22. <https://doi.org/10.1155/2022/2384830>

Pneumonia. (2025, March 19). Cleveland Clinic.

<https://my.clevelandclinic.org/health/diseases/4471-pneumonia>

OpenAI. (2024). *ChatGPT* (April 2024 version) [Large language model]. <https://chat.openai.com>

Grammarly Inc. (2024). *Grammarly writing assistant*. <https://www.grammarly.com>